Purna Chandra Sekhar Vakudavathu
Dual Degree (B.Tech. and M.Tech.)
Mathematics and Computer Science
Indian Institute of Technology, Delhi

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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
Dual Degree (Math and Computing.)	Indian Institute of Technology, Delhi	7.03	2021-Present
B.Tech. Computer Science	National Institute of Technology, Warangal	7.79	2020-2021
Senior Secondary	CBSE Board	88.2%	2020
Secondary	Telangana State Board	10	2018

Experience

- **Open Science and Accessibility Steering Group (OScA)** *December 2021 - present*
  iGEM Community Cambridge, MA, USA
  – Led the team of the Open Talks project in formulating the Open Science Training project of OScA 2022.
  – Aided the team in organizing various Open Science activities for iGEM 2022 season.

Relevant Projects

- **Advanced Graph Path Searching** *November 2022*
  Prof. Ashish Chiplunkar
  – Developed a modified version of Dijkstra's Algorithm in python to find the path of maximum capacity.
  – Adopted the priority queue data structure to develop an efficient algorithm.
- **Approximate Pattern-Searching Algorithm** *October 2022*
  Prof. Naveen Garg
  – Designed a time and space-efficient pattern-searching algorithm by modifying the Rabin-Karp Algorithm.
- **Simulation of Colliding Balls** *September 2022*
  Prof. Naveen Garg
  – Created a python program to simulate a time-ordered sequence of particle collisions using a priority queue.
  – Modification of the program can compute the digits of the mathematical constant *pi*.
- **Modified Vending Cart Design** *July 2022 - November 2022*
  RuTAG IIT Delhi, Prof. Subir Kumar Saha
  – Aided in designing a CAD-based prototype of a low-cost vending cart suitable for an Indian Demographic.
- **Python Compiler and Boolean Logic Circuits** *February 2022, December 2021*
  Prof. Preeti Ranjan Panda and Prof. Rahul Narain
  – Created a python program to read the input text and compile the basic python syntax, such as arithmetic statements, conditional statements, and loops.
  – Implemented full and half adder circuits, Boolean multiplication, and subtraction operations in python.
- **Biologically Engineered Organism for the reduction of Gossypol** *February 2021 - November 2021*
  iGEM NIT Warangal, Prof. Ashish A Prabhu
  – Screened various organisms and expressed the suitable reducing organism on a Pichia Pastoris expression vector.
  – Published a review paper on the Biological Detoxification of Gossypol in the MSP Vector Journal of iGEM 2021.
  – Facilitated the Human Practices, Education, Collaborations, and Outreach activities of the NITW Team.

Technical Skills

- **Programming and Web**: Python, C/C++, MATLAB*, R*, LaTeX*, HTML*
- **Tools and OS**: Autodesk Inventor, MS Office Suite, Windows, Linux* ** Elementary proficiency*

Workshops and Conferences

- **Inclusion Fellow**, Robotics Science and Systems (RSS) *2021*
- **Quantum Computing Hard and Soft Summer School**, Jointly organized by EPFL and ETH Zurich *2021*
- **Research Paper Writing and Scientific Presentation**, Wise Up Communications *2021*

Achievements and Positions of Responsibility

- **Silver Medal**, iGEM Giant Jamboree *2021*
- **Member**, RuTAG, IIT Delhi *March 2022 - Present*
- **Member**, iGEM, NIT Warangal *February 2021 - November 2021*
- **Member**, Business Club, NIT Warangal *January 2021 - November 2021*